Freelance Marketplace Platform

ASSIGNMENT 2 (SP-SRS)

Module 5: Collaboration & Team Workspace

Software Requirements Specification (SRS)

Section: SE - B

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Subject: Software Project Management

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1. Introduction

1.1 Purpose

This document presents the Software Requirements Specification (SRS) for the Collaboration & Team Workspace Module (Module 5) of the Freelance Marketplace Platform. The purpose of this document is to comprehensively define the functional and non-functional requirements of the module so that team formation, role-based access control, task management, and workspace functionality are clearly specified for all stakeholders.

This SRS serves as a reference for developers, testers, system integrators, and project managers involved in the design, development, and validation of Module 5.

1.2 Document Conventions

The following conventions are used throughout this document:

- Functional requirements are identified with the prefix REQ-CTW (Collaboration & Team Workspace).
- Requirements are written using the word "shall" to indicate mandatory system behaviour.
- Priority levels are defined as: High, Medium, and Low.
- RBAC refers to Role-Based Access Control.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- System Developers who will implement the Collaboration & Team Workspace module.
- Integration Team responsible for connecting all modules of the Freelance Marketplace Platform.
- Test Engineers responsible for validating system performance and functionality.
- Project Managers overseeing project delivery and scope management.
- Clients and Freelancers who represent the end-user population.

1.4 Module Scope

The Collaboration & Team Workspace Module provides a structured, project-based environment for freelancers and clients to collaborate effectively. The module encompasses:

- Team formation and member invitation

- Role-based access control (RBAC) for workspace participants
- Task creation, assignment, and progress tracking
- Workspace dashboard displaying project progress and team activity
- Activity logging of all significant workspace events

2. Overall Description

2.1 Module Perspective

The Collaboration & Team Workspace Module is a core subsystem within the Freelance Marketplace Platform. It operates within a larger ecosystem and depends on several other platform modules to deliver its functionality. The module interacts with the following subsystems:

- User Authentication & Profile Management : for session validation and identity verification
- Project & Contract Management : for project context and contract binding
- Notification System : for alerting users about task updates and invitations
- Payment & Billing Module : for milestone-based payment triggers linked to task completion

The module manages the collaboration lifecycle within a project: team setup, role enforcement, and task coordination.

2.2 Module Functions

The Collaboration & Team Workspace Module provides the following high-level functions:

- Create and manage team workspaces scoped to a specific project
- Invite members and assign roles with corresponding access permissions
- Create, assign, update, and track tasks within the project
- Display a unified workspace dashboard with project progress metrics
- Track and display workspace activity logs

2.3 User Classes and Characteristics

2.3.1 Client (Project Owner)

Clients are individuals or organizations that create projects on the platform and engage freelancers. Within the workspace, a client typically acts as the project owner with the highest level of access.

- Can create and configure the team workspace

- Can invite freelancers and assign roles
- Can create, assign, and close tasks
- Can view all workspace activity
- Moderate technical knowledge expected

2.3.2 Freelancer (Team Member)

Freelancers are independent professionals assigned to work on a project. They operate within the workspace according to the role assigned to them.

- Can view and update tasks assigned to them
- Access is governed by RBAC rules set by the project owner

2.3.3 Project Manager (Lead Freelancer / Team Lead)

A Project Manager is a freelancer granted elevated permissions to coordinate the team on behalf of the client.

- Can create and assign tasks to other team members
- Can monitor team progress and activity logs
- Cannot modify workspace-level settings or billing

2.3.4 System Administrator

System Administrators have platform-wide access and are responsible for maintaining system integrity. They do not participate in individual workspaces but can access logs and enforce platform policies.

2.4 Design and Implementation Constraints

- The module shall integrate with other platform modules through well-defined RESTful APIs.
- All workspace access shall be enforced through the RBAC system; no direct database access is permitted by end users.
- The module shall comply with platform-wide data privacy and security policies.
- The module shall adhere to standardized authentication mechanisms used across the platform (JWT or equivalent).

2.5 Assumptions and Dependencies

Assumptions

- Users are authenticated before accessing any workspace feature.
- A project must exist and be in an active contract state before a workspace can be initialized.

Dependencies

- Authentication Module : session management and user identity
- Notification Module : delivery of task alerts and member invitations
- Project & Contract Module : project ID and contract status validation

3. External Interface Requirements

3.1 User Interfaces

The module shall provide the following user-facing interface components:

- Workspace Dashboard : displays project overview, team members, progress percentage, and recent activity
- Team Management Panel : interface for inviting members, assigning roles, and viewing the team roster
- Task Board : Kanban-style or list-based view for creating, assigning, and tracking tasks
- Activity Log Viewer : chronological log of workspace events accessible to authorized roles

3.2 System Module Interfaces

The Collaboration & Team Workspace Module communicates with internal platform modules via the following interfaces:

- Authentication API : validates user sessions and permissions before granting workspace access
- Notification API : triggers alerts for task updates and member invitations
- Project/Contract API : retrieves and validates project ID and active contract status

4. Module Features

4.1 Team Formation & Member Invitation

4.1.1 Description and Priority

This feature enables a client (project owner) to create a team workspace, invite freelancers via user ID or email, and build a team scoped to a specific project.

Priority: **High**

4.1.2 Stimulus / Response Sequences

1. Client creates or opens an active project on the platform.
2. Client navigates to the Workspace section and initializes the team workspace.
3. Client enters the email addresses or user IDs of freelancers to invite.
4. System validates that the invited users exist on the platform.
5. System sends invitation notifications to the invited users.
6. Invited users accept the invitation and are added to the workspace.
7. System confirms team membership and updates the team roster.

4.1.3 Functional Requirements

REQ-CTW-01

The system shall allow the project owner (client) to create a team workspace associated with a specific project ID.

REQ-CTW-02

The system shall allow the project owner to invite users by email address or platform user ID.

REQ-CTW-03

The system shall validate that invited users have active accounts on the platform before sending invitations.

REQ-CTW-04

The system shall send an invitation notification to each invited user via the Notification Module.

REQ-CTW-05

The system shall add a user to the workspace upon acceptance of the invitation.

REQ-CTW-06

The system shall prevent duplicate membership; a user may only be a member of a given workspace once.

REQ-CTW-07

The system shall allow the project owner to remove a member from the workspace.

4.2 Role-Based Access Control (RBAC)

4.2.1 Description and Priority

This feature enforces structured access permissions within the workspace based on the role assigned to each member. Roles determine what actions a user can perform.

Priority: **High**

4.2.2 Stimulus / Response Sequences

8. Project owner assigns a role to each team member at the time of invitation or subsequently.
9. Member attempts to perform an action (e.g., create a task).
10. System checks the member's assigned role against the permission matrix.
11. System grants access if the role has the required permission, or returns an error if not.

4.2.3 Functional Requirements

REQ-CTW-08

The system shall support the following roles within a workspace: Project Owner, Project Manager, and Team Member.

REQ-CTW-09

The system shall enforce a permission matrix that governs which actions each role may perform within the workspace.

REQ-CTW-10

The system shall allow the Project Owner to assign and modify roles of workspace members.

REQ-CTW-11

The system shall prevent Team Members from modifying workspace-level settings or other members' role assignments.

REQ-CTW-12

The system shall deny access and display an appropriate error message when a user attempts an action not permitted by their role.

REQ-CTW-13

The system shall apply RBAC rules to all workspace features including tasks and activity logs.

4.3 Task Management

4.3.1 Description and Priority

This feature allows authorized users to create, assign, update, and track tasks within the project workspace. It serves as the primary tool for coordinating work among team members.

Priority: **High**

4.3.2 Stimulus / Response Sequences

12. Authorized user (Project Owner or Project Manager) navigates to the Task Board.
13. User creates a new task by providing a title, description, deadline, and assignee.
14. System saves the task and notifies the assigned team member.

15. Assignee updates the task status (e.g., In Progress, Completed).
16. System records the update and reflects the new status on the Task Board.
17. System recalculates and updates the overall project progress percentage.

4.3.3 Functional Requirements

REQ-CTW-14

The system shall allow authorized users to create a new task with a title, description, deadline, priority level, and assignee.

REQ-CTW-15

The system shall associate each task with a project ID, workspace ID, creator ID, and assignee ID.

REQ-CTW-16

The system shall record a timestamp upon task creation and upon each subsequent status update.

REQ-CTW-17

The system shall support the following task statuses: To Do, In Progress, Under Review, and Completed.

REQ-CTW-18

The system shall notify the assigned team member when a task is created or reassigned via the Notification Module.

REQ-CTW-19

The system shall allow the assignee to update the status of a task assigned to them.

REQ-CTW-20

The system shall allow authorized users to edit task details including title, description, deadline, and assignee.

REQ-CTW-21

The system shall allow authorized users to delete a task from the workspace.

REQ-CTW-22

The system shall calculate and display the overall project progress as a percentage based on the ratio of completed tasks to total tasks.

REQ-CTW-23

The system shall display a task board listing all tasks with their current status, assignee, and deadline.

4.4 Activity Logging & Workspace Dashboard

4.4.1 Description and Priority

This feature maintains a comprehensive log of all significant actions within the workspace and presents an integrated dashboard summarizing project status, team membership, and progress.

Priority: **High**

4.4.2 Functional Requirements

REQ-CTW-24

The system shall log all significant workspace actions including member additions/removals, task creation/updates, and role changes.

REQ-CTW-25

Each activity log entry shall include the action type, the user who performed it, and a timestamp.

REQ-CTW-26

The system shall display the activity log in reverse chronological order.

REQ-CTW-27

The system shall restrict full activity log access to the Project Owner and Project Manager roles.

REQ-CTW-28

The system shall display a workspace dashboard showing: project name, team member list, overall task progress percentage, and recent activity.

REQ-CTW-29

The workspace dashboard shall reflect the latest project state without requiring a manual page refresh.

5. Non-Functional Requirements

5.1 Performance Requirements

- The workspace dashboard shall load and render fully within 3 seconds under normal network conditions.
- The task board shall retrieve and display up to 200 tasks without noticeable latency.
- The system shall support at least 100 concurrent workspace sessions without performance degradation.

5.2 Security Requirements

- All API communications between the module and external systems shall use HTTPS with TLS 1.2 or higher.
- Users shall only be able to access workspaces they are members of; cross-workspace data access shall be strictly prohibited.
- Role-based access control checks shall be enforced server-side and shall not rely solely on client-side validation.
- All sensitive user data transmitted through the module shall be encrypted in transit.

5.3 Software Quality Attributes

Usability

The workspace interface shall be intuitive and consistent, enabling users with basic technical knowledge to navigate the task board and activity log without requiring a tutorial.

Reliability

The module shall maintain a minimum uptime of 99.5%. Task state shall be preserved and consistent at all times.

Scalability

The module architecture shall support horizontal scaling to accommodate an increasing number of concurrent workspaces and users as the platform grows.

Maintainability

The module shall be designed with clearly separated components (Team Manager, Task Manager, Role Manager, Activity Logger, Workspace UI) to facilitate independent maintenance and updates.

Portability

The module shall function correctly across major modern web browsers (Chrome, Firefox, Edge, Safari) and shall be responsive for desktop and tablet screen sizes.